

Hydrops–Lactic Acidosis–Sideroblastic Anemia–Multisystemic Failure Syndrome (HLASA)

Comprehensive Disease Characterization Report

MONDO:0014869 · Gene: *LARS2* · Category: Mendelian (autosomal recessive mitochondrial disorder)

Evidence base: This is an ultra-rare disorder with fewer than ~10 reported patients worldwide. The knowledge base derives almost entirely from **individual patient case reports** (not aggregated registries): the index proband (Riley et al., 2016,

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and three additional cases from two families (Riley et al., 2020,

P 32442335

), supplemented by the wider *LARS2*/Perrault-syndrome and congenital sideroblastic anemia literature. Where disease-level facts are unavailable, this is stated explicitly.

1. Disease Information

Overview. HLASA is an infantile-lethal, autosomal-recessive **mitochondrial multisystem metabolic disorder** caused by biallelic pathogenic variants in *LARS2*, the nuclear gene encoding **mitochondrial leucyl-tRNA synthetase**. It represents the **most severe end of the *LARS2* phenotypic spectrum**, which ranges (mild → severe) from Perrault syndrome (deafness + primary ovarian insufficiency), through deafness with (ovario-)leukodystrophy and reversible mitochondrial myopathy, to lethal neonatal HLASA. The cardinal features are non-immune **hydrops fetalis**, severe **lactic acidosis**, and **sideroblastic anemia**, accompanied by multiorgan (cardiac, pulmonary, renal, hepatic) failure.

"In this study, we report variants in *LARS2* that are associated with a severe multisystem metabolic disorder. The proband was born prematurely with severe lactic acidosis, hydrops, and sideroblastic anemia." —

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Key identifiers. - **MONDO:** MONDO:0014869 - **Gene:** *LARS2*, **HGNC:**21353, **OMIM** *604544, locus 3p21.3, **UniProt** Q15031 - **Related OMIM phenotype:** Perrault syndrome 4, #615300 (allelic series) - **Orphanet:** Perrault syndrome (ORPHA:2855) is the closest catalogued entity; HLASA is described as a *LARS2*-related severe variant. - **ICD-11:** best fit 5C53 / 5C50.4 (mitochondrial/metabolic disorders); **ICD-10:** E88.8 (other specified metabolic disorders) / D64.0 (hereditary sideroblastic anemia) — no dedicated code. - **MeSH:** no specific term; indexed under "Anemia, Sideroblastic," "Mitochondrial Diseases," "Hydrops Fetalis."

Synonyms / alternative names. HLASA; Hydrops, Lactic Acidosis, and Sideroblastic Anemia; *LARS2*-related hydrops-lactic acidosis-sideroblastic anemia and multisystem failure.

2. Etiology

Primary cause — genetic. Biallelic (compound heterozygous in all reported families) pathogenic variants in *LARS2*. Index proband: **c.1289C>T (p.Ala430Val)** and **c.1565C>A (p.Thr522Asn)**; both predicted damaging (SIFT, PolyPhen). p.Thr522Asn was previously reported in Perrault syndrome.

"Whole exome sequencing of patient DNA revealed compound heterozygous variants in *LARS2* (c.1289C>T; p.Ala430Val and c.1565C>A; p.Thr522Asn)." —

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Genetic risk factors. The disease is monogenic/Mendelian; the only "risk factor" is inheriting two damaging *LARS2* alleles. Being a **carrier** (heterozygote) is not associated with disease. **Consanguinity/shared ancestry** increases recurrence risk (as for all AR disorders). No

validated modifier genes are established, though residual aminoacylation activity of the specific allele combination is the principal determinant of severity (genotype–phenotype correlation, see §4/§6).

Environmental / infectious factors. None. HLASA is **not** caused or triggered by toxins, infection, or lifestyle. Intercurrent illness and physiologic stress (e.g., prematurity, pregnancy in milder *LARS2*/*YARS2*-related disease) can precipitate metabolic decompensation but are not causal.

Protective factors. None described genetically or environmentally. (Not applicable for an ultra-rare Mendelian lethal disorder.)

Gene–environment interactions. Not established. Metabolic stress may unmask/aggravate mitochondrial insufficiency (inferred from the broader mt-aaRS literature), but no specific GxE data exist for HLASA.

3. Phenotypes

All are **congenital/neonatal onset**, **severe**, and **rapidly progressive** in classic HLASA; two of four reported patients survived the neonatal period with residual chronic morbidity (developmental delay, deafness). Frequencies below are from the ≤ 4 reported HLASA cases (small-N; qualitative).

Phenotype	Type	HPO term	Onset/ severity	Frequency (reported cases)
Non-immune hydrops fetalis	physical/ clinical sign	HP:0001789	prenatal/ neonatal, severe	core feature
Lactic acidosis	lab abnormality	HP:0003128	neonatal, severe	core feature
Sideroblastic anemia (ring sideroblasts)	lab abnormality	HP:0001924	neonatal, severe	core feature
Pulmonary hypertension	clinical sign	HP:0002092	neonatal	index case
Hyaline membrane disease / RDS	clinical sign	HP:0002094 (resp. insufficiency)	neonatal	index case
Impaired cardiac function	clinical sign	HP:0001637 / HP:0001635	neonatal	index case
Coagulopathy	lab abnormality	HP:0001928	neonatal	index case
Progressive renal disease	clinical sign	HP:0000112	neonatal	index case
Prematurity	clinical	HP:0001622	perinatal	index case
Male genital anomalies / undervirilization (incl. hypospadias, cryptorchidism)	physical	HP:0000811 / HP:0000047 / HP:0000028	congenital	all survivors (males)
Sensorineural hearing loss	clinical sign	HP:0000407	infancy	survivors
Global developmental delay	clinical sign	HP:0001263	infancy	survivors

"She had multisystem complications with hyaline membrane disease, impaired cardiac function, a coagulopathy, pulmonary hypertension, and progressive renal disease and succumbed at 5 days of age." —

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"All were males with genital anomalies. Two survived multisystem disease in the neonatal period; both have developmental delay and hearing loss." —

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Quality of life. In lethal cases QoL assessment is not applicable (death at ~5 days). Survivors face lifelong burden from deafness and neurodevelopmental disability requiring multidisciplinary support. No formal EQ-5D/SF-36/PROMIS data exist for HLASA.

4. Genetic / Molecular Information

Causal gene. *LARS2* (mitochondrial leucyl-tRNA synthetase). HGNC:21353; OMIM *604544; 3p21.3; UniProt Q15031; a **class I aminoacyl-tRNA synthetase** that charges L-leucine onto mt-tRNA^{Leu} (two isoacceptors, UUR and CUN).

Pathogenic variants (reported/HLASA-relevant). - **c.1289C>T (p.Ala430Val)** — missense; **loss-of-function** (severely reduced aminoacylation, ~18-fold decreased efficiency in vitro); HLASA index allele. - **c.1565C>A (p.Thr522Asn)** — missense; previously Perrault-associated; HLASA index allele. - Additional HLASA-associated *LARS2* missense alleles in the 2020 series produced the **most severe aminoacylation deficit** of all *LARS2* variants tested.

Variant class: predominantly **missense**; **germline**; ACMG classification pathogenic/likely pathogenic (supported by functional aminoacylation assays, in-silico prediction, segregation, rarity). No somatic contribution (contrast SF3B1 in acquired MDS-RS).

Allele frequency. Reported HLASA alleles are **rare/absent** in gnomAD (consistent with a severe recessive disorder); exact per-variant frequencies not reported in the primary papers. Carrier frequency in the general population is not established.

Functional consequence: loss of function (hypomorphic reduction of leucyl-tRNA aminoacylation). Not gain-of-function or dominant-negative. Genotype-phenotype rule: **severity is inversely proportional to residual aminoacylation activity**.

"Analysis of recombinant LARS2 variant proteins showed they had reduced aminoacylation efficiency, with HLASA-associated variants having the most severe effect."

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Modifier genes / epigenetics / chromosomal abnormalities. None established. No epigenetic mechanism or large-scale chromosomal rearrangement is implicated; the disorder is a point-mutation, single-gene condition.

5. Environmental Information

Not applicable. HLASA is a purely genetic Mendelian disorder. No environmental toxins, radiation, occupational exposures, lifestyle factors, or infectious agents cause or trigger it. (For completeness: acquired/secondary sideroblastic anemias can be caused by alcohol, isoniazid, chloramphenicol, linezolid, zinc-induced copper deficiency — but these are *not* relevant to LARS2-HLASA.)

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function LARS2 variants** are inherited → **results in** a hypomorphic mitochondrial leucyl-tRNA synthetase enzyme.
2. Reduced enzyme activity **leads to** deficient **aminoacylation (leucine-charging) of mitochondrial tRNA^{Leu}** (demonstrated in vitro; ~18-fold reduced for p.Ala430Val).

3. Deficient charged mt-tRNA^{Leu} **impairs mitochondrial translation** of the 13 mtDNA-encoded oxidative-phosphorylation (OXPHOS) subunits (inferred from enzyme role; supported by protein data).
4. Impaired mitochondrial translation **results in reduced assembly/levels of respiratory-chain complexes**, especially **Complex I** (demonstrated: reduced Complex I protein in patient muscle and liver).
5. Reduced OXPHOS **leads to** an energy (ATP) deficit and a compensatory shift to **anaerobic glycolysis → systemic lactic acidosis** (CHEBI:16113 lactate).
6. **Branch A — erythroid lineage:** OXPHOS/mitochondrial dysfunction in erythroblasts **disrupts mitochondrial heme biosynthesis and iron-sulfur cluster metabolism → mitochondrial iron accumulation** in perinuclear mitochondria → **ring sideroblasts** and **ineffective erythropoiesis → sideroblastic anemia** and fetal anemia (inferred by analogy to congenital sideroblastic anemia mechanisms).
7. Fetal anemia + high-output cardiac strain + generalized energy failure **results in** fluid extravasation and serous effusions → **non-immune hydrops fetalis** (inferred).
8. **Branch B — high-energy organs:** OXPHOS deficiency in heart, lung vasculature, kidney, and liver **leads to** impaired cardiac function, pulmonary hypertension, progressive renal disease, hepatic dysfunction, and coagulopathy → **multiorgan failure** and neonatal death (demonstrated clinically).
9. **Branch C — survivors / milder spectrum:** partial mitochondrial translation deficiency in cochlea and CNS **leads to** sensorineural hearing loss, developmental delay (and, at the milder Perrault end, leukodystrophy and ovarian dysfunction); in gonads it **impacts development** (male undervirilization; *C. elegans lars-2* loss abolishes germ-cell production).

Supporting detail by category

- **Molecular pathways:** mitochondrial translation / OXPHOS (KEGG hsa00190 oxidative phosphorylation; Reactome "Mitochondrial translation," "Metabolism of amino acids"). Aminoacyl-tRNA biosynthesis (KEGG hsa00970).
- **Cellular processes:** ineffective erythropoiesis, cellular energy failure, apoptosis of energy-starved cells (inferred), impaired erythroid maturation.
- **Protein dysfunction: loss of function** of a class-I aaRS via missense substitutions reducing catalytic aminoacylation efficiency (not aggregation/misfolding-driven per se).
- **Metabolic changes:** lactic acidemia; disrupted heme/iron handling; energy-metabolism failure.

- **Biochemical abnormality:** enzyme deficiency — mitochondrial leucyl-tRNA ligase (EC 6.1.1.4); secondary Complex I (NADH:ubiquinone oxidoreductase) deficiency.
- **Tissue damage mechanism:** bioenergetic insufficiency / ischemic-type energy starvation in high-demand tissues; mitochondrial iron toxicity in erythroblasts.
- **Immune involvement:** none (hydrops is non-immune).
- **Molecular profiling:** patient-tissue immunoblot showed reduced LARS2 (liver) and reduced Complex I (muscle, liver); notably **RC enzyme activities were not markedly deficient** in the index proband's muscle/liver, indicating a partial/tissue-variable biochemical lesion.

"Muscle and liver samples from the proband did not display marked mitochondrial respiratory chain enzyme deficiency... complex I protein levels were reduced in patient muscle and liver."

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GO / CL suggestions. BP: GO:0070127 (mitochondrial translation), GO:0006783 (heme biosynthetic process), GO:0032543 (mitochondrial translation), GO:0006418 (tRNA aminoacylation for protein translation), GO:0045333 (cellular respiration). MF: GO:0004823 (leucine-tRNA ligase activity). CC: GO:0005739 (mitochondrion), GO:0005759 (mitochondrial matrix). Cells: CL:0000764 (erythroblast), CL:0000232 (erythrocyte), CL:0000746 (cardiac muscle cell).

7. Anatomical Structures Affected

- **Primary organs / systems:** hematopoietic–erythroid (bone marrow, **UBERON:0002371**), cardiovascular (heart **UBERON:0000948**; pulmonary vasculature → pulmonary hypertension), respiratory (lung **UBERON:0002048**), renal (kidney **UBERON:0002113**), hepatic (liver **UBERON:0002107**).
- **Secondary / spectrum:** skeletal muscle (**UBERON:0001134**; reduced Complex I, degeneration), auditory system (cochlea/inner ear **UBERON:0001846**), CNS/white matter (brain **UBERON:0000955**; leukodystrophy at milder end), reproductive organs (testis **UBERON:0000473**; ovary **UBERON:0000992**). Generalized serosal cavities (hydrops).

- **Tissue/cell level:** erythroid precursors (CL:0000764 erythroblast), cardiomyocytes, cochlear hair cells, oocytes/germ cells; connective/serous tissue edema in hydrops.
 - **Subcellular: mitochondrion (GO:0005739)**, mitochondrial matrix (GO:0005759) — the primary locus of the molecular defect.
 - **Lateralization:** systemic/**bilateral** (hearing loss bilateral; multiorgan involvement generalized).
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8. Temporal Development

- **Onset: congenital** — prenatal (hydrops detectable in utero) to immediate neonatal; **acute** presentation at/shortly after birth.
 - **Progression: rapidly progressive** in classic HLASA → death typically within days (index proband died at **day 5**). Survivors stabilize after the neonatal crisis and follow a **chronic, static-to-slowly-evolving** course dominated by deafness and developmental delay.
 - **Disease course pattern:** neonatal catastrophic (lethal) **or** neonatal crisis followed by chronic residual disability.
 - **Duration:** self-limited by early death, or lifelong chronic in survivors.
 - **Remission:** none spontaneous; note the broader *LARS2* spectrum includes a **reversible** infantile mitochondrial myopathy phenotype (distinct from HLASA).
 - **Critical period:** the **perinatal/neonatal window** is both the period of maximal vulnerability and the only window for life-saving intensive support.
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9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic; compound heterozygous in reported families). Reflects the Perrault-syndrome gene family.
- **Penetrance/expressivity:** presumed high penetrance for the biallelic severe genotype; **highly variable expressivity/allelic heterogeneity** across the *LARS2* spectrum (same gene → Perrault vs leukodystrophy vs reversible myopathy vs HLASA), governed by residual enzyme activity.

- **Anticipation / mosaicism / founder effects:** none described (not a repeat-expansion disorder). Consanguinity increases recurrence risk generally.
- **Carrier frequency:** not established; alleles are individually very rare in gnomAD.
- **Epidemiology: ultra-rare** — HLASA reported in only ~4 individuals (2016 + 2020). Perrault syndrome overall (~15 causal genes) has ~100 reported cases. No prevalence/incidence estimate is available for HLASA specifically (Orphanet lists Perrault syndrome as <1/1,000,000).
- **Sex ratio:** all reported HLASA survivors are **male with genital anomalies**; the original lethal proband was female. Small-N precludes a true ratio. (In Perrault, ovarian phenotype is female-specific.)
- **Geographic/ethnic distribution:** no predilection identified; reported families are of varied ancestry.

10. Diagnostics

Laboratory / biochemical. - Blood gas & metabolic: **elevated blood lactate**, metabolic (lactic) acidosis, elevated lactate:pyruvate ratio (LOINC 2524-7 lactate); CSF lactate if CNS involved. - CBC: anemia; peripheral smear. - **Bone marrow aspirate with Perls' (Prussian blue) iron stain: ring sideroblasts ($\geq 15\%$, type 3)** — diagnostic of sideroblastic anemia. - Coagulation studies (coagulopathy), renal function, cardiac enzymes/BNP.

Biomarkers. Lactate (CHEBI:16113); ring sideroblasts; reduced Complex I protein on muscle/liver immunoblot; reduced LARS2 protein. No validated circulating molecular biomarker specific to HLASA.

Imaging / functional. Fetal/neonatal ultrasound (hydrops, effusions); echocardiography (cardiac dysfunction, pulmonary hypertension); brain MRI (leukodystrophy in milder spectrum). Audiology (ABR/OAE) for hearing loss in survivors.

Biopsy / pathology. Bone marrow (ring sideroblasts). Muscle histology may show an "unusual form of degeneration" (2020 myopathy case). Respiratory chain enzymology may be **normal or only mildly reduced** — a diagnostic pitfall.

Genetic testing (definitive). - **First-line: trio/proband whole-exome (WES) or whole-genome sequencing (WGS)** — how all reported cases were diagnosed. - **Targeted gene panels:** mitochondrial disease panels, Perrault-syndrome panels (incl. *LARS2*, *HARS2*, *CLPP*,

TWINK, *ERAL1*, *HSD17B4*), and **congenital sideroblastic anemia panels** (*ALAS2*, *SLC25A38*, *GLRX5*, *YARS2*, *PUS1*, *ABCB7*, *SF3B1*). - **Single-gene *LARS2* testing** for cascade/known familial variants; confirm biallelic status and phase (parental testing). - **mtDNA testing / karyotype / CMA / FISH / repeat-expansion testing**: used to **exclude** differentials (e.g., Pearson syndrome mtDNA deletion, MT-ATP6 variants), not to diagnose HLASA.

Clinical criteria / differential diagnosis. No formal consensus criteria (too rare). Diagnosis = characteristic triad (hydrops + lactic acidosis + sideroblastic anemia) + biallelic *LARS2* variants. **Differential:** MLASA (MLASA1 *PUS1*, MLASA2 *YARS2*, MT-ATP6); other congenital sideroblastic anemias (*ALAS2* X-linked, *SLC25A38* AR, *GLRX5*, *ABCB7*); Pearson marrow-pancreas syndrome (mtDNA deletion); other causes of non-immune hydrops; other mt-aaRS disorders.

"Our study confirms that MLASA must be considered in patients with congenital sideroblastic anemia and underlines the importance of early diagnosis and supportive therapy."

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Screening. No newborn-screening assay detects HLASA. **Carrier and cascade screening** is possible once a familial variant is known; **prenatal diagnosis / PGT** feasible for at-risk couples.

11. Outcome / Prognosis

- **Survival / mortality: poor.** Classic HLASA is **infantile-lethal** (index proband died day 5). However, **2 of 4** reported patients **survived** the neonatal multisystem crisis with intensive support — so the phenotype is not uniformly lethal.
- **Life expectancy:** days in classic lethal presentation; survivors have undefined long-term expectancy with chronic morbidity.
- **Morbidity in survivors:** developmental delay and sensorineural hearing loss (lifelong disability); genital anomalies requiring surgery.
- **Complications:** respiratory failure (hyaline membrane disease), cardiac failure, pulmonary hypertension, coagulopathy, renal failure, transfusion-dependent anemia and iron overload.

- **Recovery potential:** no reversal of the underlying enzyme defect; survivors do not "recover" but can stabilize.
 - **Prognostic factors:** degree of residual *LARS2* aminoacylation activity (genotype), severity of neonatal acidosis and multiorgan involvement, and availability of intensive neonatal support. **Prognostic biomarker:** persistently high lactate reflects ongoing bioenergetic failure.
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12. Treatment

No approved disease-specific/curative therapy exists. Management is **supportive and multidisciplinary** (NCIT: Supportive Care; Palliative Care).

- **Sideroblastic anemia:** empiric **pyridoxine (vitamin B6)** trial (NCIT Pyridoxine; CHEBI:27306) — responsiveness variable and generally limited in mitochondrial-translation forms; **packed red-cell transfusion** (NCIT Red Blood Cell Transfusion) for transfusion dependence; **iron chelation** (deferasirox/deferoxamine; NCIT Iron Chelation Therapy) for iron overload. Fetal anemia/hydrops from mitochondrial disease has been managed with **intrauterine transfusion** in related cases.
- **Metabolic/acidosis:** correction of acidosis, glucose/energy support, avoidance of mitochondrial toxins.
- **Organ support:** mechanical ventilation (RDS), inotropes and pulmonary-hypertension therapy, renal support, correction of coagulopathy.
- **Mitochondrial "cocktails":** empiric antioxidants/cofactors/vitamins (CoQ10, riboflavin, thiamine, L-carnitine, etc.) are used without validated efficacy. A **preclinical** study showed synergistic rescue with **glucose + nicotinic acid + N-acetylcysteine** in *C. elegans* and zebrafish **Complex I** disease models — mechanistically relevant but **not clinically validated**.
- **Survivor care:** hearing habilitation (hearing aids/cochlear implants), developmental/rehabilitation therapies (PT/OT/speech), urologic surgery for genital anomalies.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno): none available;** no HLASA-specific clinical trials (ClinicalTrials.gov). HSCT is curative only for select *other* congenital sideroblastic anemias (e.g., *SLC25A38*), not for *LARS2*-HLASA.
- **Pharmacogenomics:** not applicable.

"Synergistic rescue occurred only with glucose, nicotinic acid and N-acetylcysteine (Glu + NA + NAC), yielding improved mitochondrial membrane potential..." —

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(preclinical)

13. Prevention

- **Primary prevention:** not possible (genetic). **Genetic counseling** for at-risk couples; **carrier screening** where a familial variant is known.
- **Reproductive options:** **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** for couples with two known *LARS2* variants; prenatal ultrasound surveillance for hydrops in at-risk pregnancies.
- **Secondary prevention:** early molecular diagnosis to guide anticipatory management (transfusion, acidosis control) and prognostic counseling.
- **Tertiary prevention:** prevent complications in survivors — iron-overload monitoring/chelation, audiologic and developmental surveillance, cardiac/renal monitoring.
- **Immunization / public health / behavioral / environmental interventions:** not applicable (no environmental etiology).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *LARS2* is evolutionarily conserved. Orthologs: mouse *Lars2* (NCBITaxon:10090), zebrafish *lars2* (NCBITaxon:7955), *Caenorhabditis elegans* **lars-2** (NCBITaxon:6239), *S. cerevisiae* mitochondrial LeuRS *NAM2* (NCBITaxon:4932).
- **Natural disease in animals:** no naturally occurring HLASA-equivalent disease reported in companion animals or wildlife (OMIA: none catalogued). No veterinary or zoonotic relevance.
- **Comparative biology:** the requirement of mitochondrial LeuRS for OXPHOS translation is conserved across eukaryotes; *C. elegans* *lars-2* loss abolishes germ-cell production, mirroring the gonadal/reproductive phenotype seen in human *LARS2* disease.

"...complete infertility due to failure to produce germ cells in *Caenorhabditis elegans* (*C. elegans*), indicating that *LARS2* is expressed in gonadal tissue and can impact gonadal development."

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15. Model Organisms

- **Invertebrate: *C. elegans lars-2*** — informative for gonadal/germ-cell role; general mitochondrial Complex I disease worm models (e.g., *gas-1(fc21)/ndufs2*) used for therapeutic screening

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- **Vertebrate: zebrafish** Complex I disease/inhibition models (rotenone;

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recapitulate lactate elevation, reduced activity, and neuro-metabolic failure — a platform for testing metabolic rescue.

- **In vitro / biochemical: recombinant *LARS2* aminoacylation assays** (the key functional model establishing variant pathogenicity and genotype–severity correlation; PMIDs 26537577, 32442335); patient-derived muscle/liver samples.
- **Genetic models available:** knockdown/knockout in worm and fish; **no published *Lars2* HLASA-specific mouse knock-in** recapitulating the full syndrome.
- **Phenotype recapitulation & limitations:** cellular/invertebrate models capture the **bioenergetic (Complex I/OXPHOS) deficit** and enable drug screening but **do not reproduce the full multisystem neonatal HLASA phenotype** (hydrops, sideroblastic anemia, human-specific developmental features). This is the principal gap for mechanistic and preclinical therapeutic work.
- **Resources:** WormBase (*lars-2*), ZFIN (*lars2*), MGI (*Lars2*), Alliance of Genome Resources.

Evidence Summary & Source Types

- **Human clinical (case reports):**

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(index proband, functional assays),

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(spectrum incl. 3 HLASA cases + recombinant protein assays),

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(LARS2 Perrault + gonadal/undervirilization + C. elegans),

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(Perrault gene family/AR inheritance),

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(mitochondrial fetal anemia/IUT),

P 35393742

& 25638461 (YARS2/MLASA differential).

- **In vitro / biochemical:** recombinant aminoacylation assays (26537577, 32442335).

- **Model organism / preclinical:**

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(C. elegans + zebrafish Complex I therapy).

- **Reviews:**

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(mt-aaRS disorders),

P 32759564

39358290 / 40769913 (sideroblastic anemia pathophysiology),

P 26637696

(SA diagnosis/treatment).

Limitations

- Extremely small evidence base (≤ 4 HLASA patients); frequencies, prevalence, sex ratio, and prognosis are imprecise.
- No HLASA-specific animal model; branch-point mechanisms for sideroblastic anemia and hydrops are **inferred** from congenital sideroblastic anemia biology, not directly demonstrated in HLASA tissue.
- No clinical trials or disease-modifying therapies.

Future Directions

- Generate a faithful *Lars2* knock-in/erythroid-conditional mammalian model to test the erythroid heme/Fe-S branch and candidate therapies.
- Systematic genotype–residual-activity–phenotype mapping across *LARS2* alleles.
- Evaluate mitochondrial cofactor combinations (e.g., Glu+NA+NAC) and mt-tRNA-directed strategies in HLASA-relevant systems.

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